COMSATS University Islamabad – Abbottabad Campus

Department of Computer Engineering

Signals and Systems Semester Project

Speech Signal Recording, Spectral Analysis and Basic Voice Characterization Using MATLAB GUI

Course: Signals and Systems **Semester:** 4th Semester

Course Code: CPE223 **Section:BCE-4A**

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1)Abstract:

Speech signal processing is an important application of Digital Signal Processing (DSP). This project presents a MATLAB-based Graphical User Interface (GUI) that records speech using a microphone, plays the recorded signal, displays its time-domain waveform, performs Fast Fourier Transform (FFT) for frequency-domain analysis, removes noise using digital filtering, and estimates the speaker's pitch to classify the voice as male or female using threshold-based logic. The GUI provides an interactive environment to understand the fundamental concepts of speech signal analysis.

2)Introduction:

Speech signals contain important information in both the time domain and frequency domain. MATLAB provides powerful signal processing functions that make speech analysis easier. The objective of this project is to design a user-friendly MATLAB App Designer GUI that performs speech recording, visualization, filtering, spectral analysis and simple voice characterization.

3)Objectives:

The objectives of this project are:

- Record speech using a microphone.
- Play recorded speech.
- Play only the selected duration of speech.
- Plot speech signal in the time domain.
- Perform FFT-based frequency spectrum analysis.
- Reduce background noise using a digital low-pass filter.
- Play filtered speech.
- Estimate pitch using autocorrelation.
- Classify speaker as Male or Female.

4)Software Requirements:

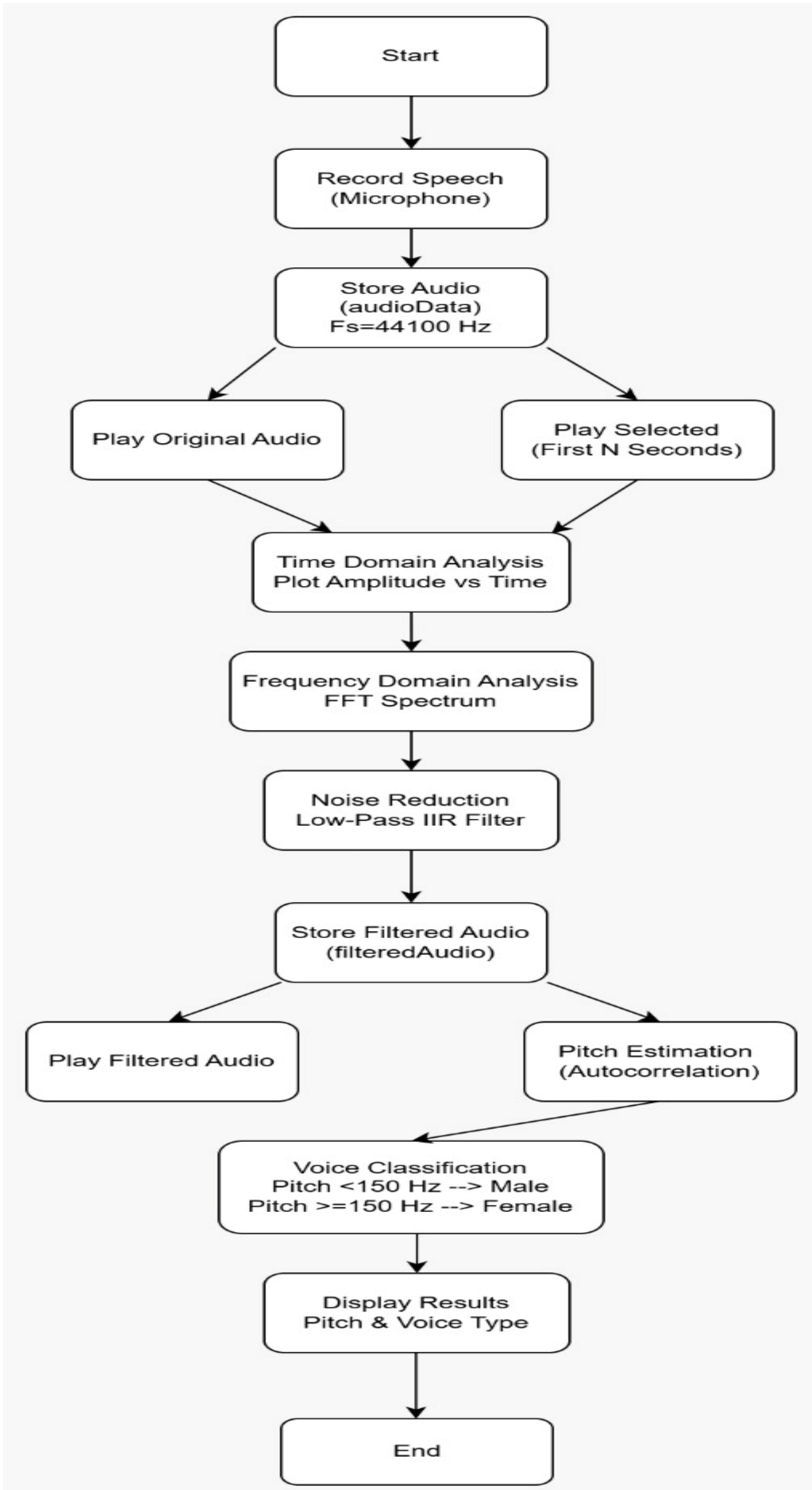
- MATLAB Online
- MATLAB App Designer
- MATLAB Signal Processing Functions
- Computer with Microphone

5)GUI Components:

The developed GUI contains the following components:

Component	Function
Record Button	Records speech
Play Original	Plays original recording
Play Selected	Plays first N seconds
Plot Signal	Displays time-domain waveform
FFT Spectrum	Displays frequency spectrum
Remove Noise	Applies digital filtering
Play Filtered	Plays filtered audio
Detect Voice	Estimates pitch and classifies voice
Time Domain Tab	Displays waveform
Frequency Domain Tab	Displays FFT graph

6)BLOCK DIAGRAM:



7). Methodology:

The proposed system is developed using MATLAB App Designer to perform speech signal recording, analysis, filtering, and voice characterization. The overall working methodology consists of the following stages.

Step 1: Audio Recording:

The application records the user's speech through the system microphone using MATLAB's audiorecorder function. The recording duration can be specified by the user (default 8 seconds). The recorded speech signal is sampled at **44.1 kHz (44100 Hz)** and stored in the variable audioData for further processing.

Step 2: Audio Playback:

After recording, the user can listen to the complete speech signal using the **Play Original** button. The application also provides a **Play Selected** option, which allows playback of only the first **N seconds** of the recorded speech based on the value entered by the user.

Step 3: Time-Domain Analysis:

The recorded speech signal is displayed in the time domain by plotting **amplitude versus time**. This visualization helps observe variations in the speech waveform and provides a better understanding of the temporal characteristics of the signal.

Step 4: Frequency-Domain Analysis:

To analyze the frequency content of the speech signal, the Fast Fourier Transform (FFT) is applied. The resulting frequency spectrum is plotted over a frequency range of **0–5000 Hz**, allowing the user to identify the dominant frequency components present in the speech.

Step 5: Noise Reduction:

The application applies an **8th-order Low-Pass IIR digital filter** with a passband frequency of **3000 Hz** to reduce unwanted high-frequency noise. The filtered speech signal is stored separately in filteredAudio, ensuring that the original recording remains unchanged. The filtered signal can also be played using the **Play Filtered** button.

Step 6: Pitch Estimation and Voice Characterization:

The filtered speech signal is processed using the **autocorrelation method** to estimate the fundamental frequency (pitch).

The estimated pitch is then compared with a predefined threshold to classify the speaker's voice:

- **Pitch < 150 Hz → Male**
- **Pitch ≥ 150 Hz → Female**

Finally, the application displays both the **estimated pitch value** and the **detected voice type** on the GUI.

8. Results

GUI Home Screen

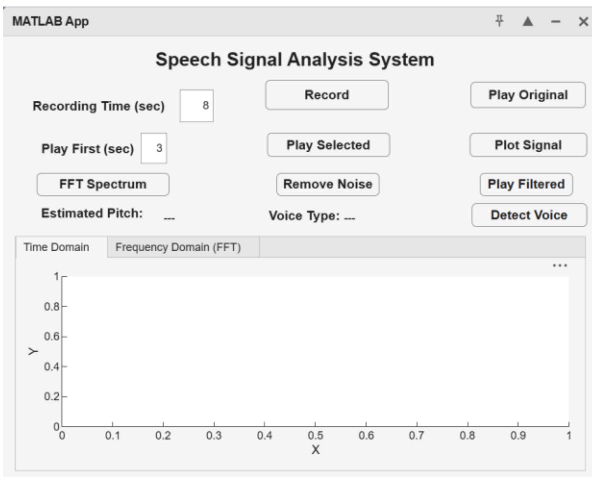


Figure 1

Time Domain Waveform

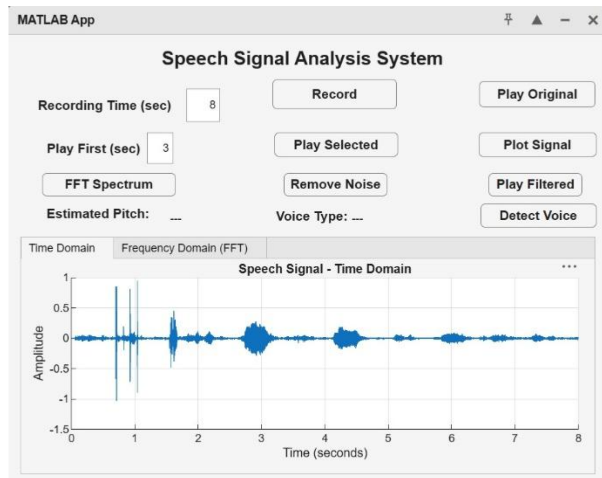


Figure 2

FFT Spectrum

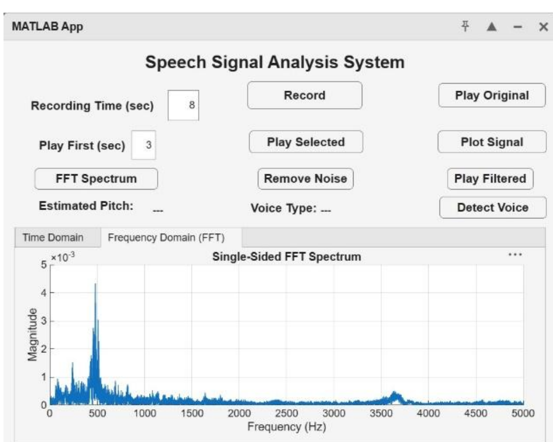


Figure 3

Voice Detection Result

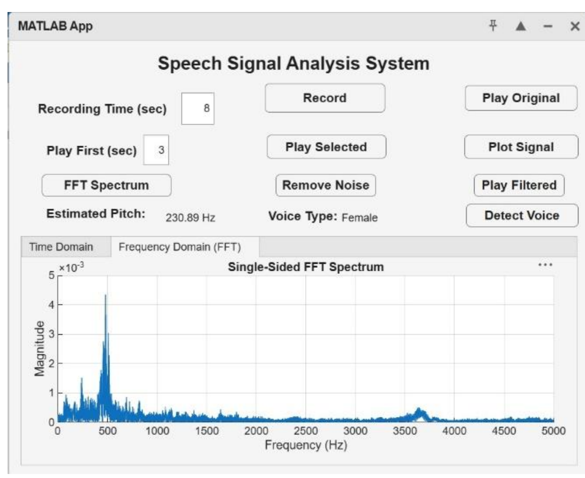


Figure 4

9)Learning Outcomes:

During this project the following concepts were learned:

- Speech Signal Processing
- MATLAB GUI Development
- Audio Recording Techniques
- Time Domain Analysis
- Frequency Domain Analysis
- Fast Fourier Transform
- Digital Filtering
- Pitch Estimation
- Voice Classification
- Signal Visualization

10)Conclusion:

A MATLAB-based speech signal analysis system was successfully developed using App Designer. The application provides an interactive interface for recording speech, visualizing waveforms, performing FFT-based frequency analysis, reducing noise using digital filters, and estimating pitch for basic voice classification. The project demonstrates the practical implementation of Digital Signal Processing concepts and fulfills all the required objectives of the semester project.

11)Future Improvements:

The system can be enhanced by adding:

- Speech-to-Text Conversion
- Automatic Speech Recognition
- Real-Time Audio Processing
- Spectrogram Display
- Save Recorded Audio (.wav)
- Load Existing Audio Files
- Machine Learning-based Speaker Recognition
- Emotion Detection
- More Accurate Pitch Detection Algorithms (YIN or Cepstrum)